

From Metawebs to Realised Food Webs: A Process-Based Framework for Network Representation Across Ecological Scales

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Abstract: Ecological networks are increasingly used to understand biodiversity and predict ecosystem responses to environmental change, yet different network representations encode different assumptions about how trophic interactions arise and what ecological processes they represent. Here we present a process-based framework for trophic ecological networks that views food web construction as a hierarchical transition from metawebs, which represent the feasibility of feeding interactions, to realised food webs, which represent interactions expressed within specific spatial and temporal community contexts. We identify five key processes underlying this transition: evolutionary compatibility and co-occurrence, which define interaction feasibility, and abundance, diet choice, and non-trophic interactions, which determine interaction realisation. We further position these processes along a continuum of network construction methods, from inductive approaches that infer structure from observations to deductive approaches that generate structure from mechanistic principles. By explicitly linking network representations to their underlying assumptions and scale-dependent processes, our framework clarifies the questions that different networks can address, highlights key challenges in moving between potential and realised interactions, and provides a roadmap for developing predictive network ecology. We argue that greater conceptual clarity is essential for advancing biodiversity science and for the effective application of networks in conservation, environmental management, and forecasting ecological change in the Anthropocene.

Keywords: food web, network construction, biodiversity, scale and process, interaction modelling, trophic network

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1 Introduction

At the heart of modern biodiversity science are theories linking species richness, stability, and ecosystem functioning, from the foundational work of Elton (2001), Lindeman (1942), and MacArthur (1955) through to contemporary questions about biodiversity loss and ecosystem resilience. Trophic ecological networks (food webs) have become a central representation for studying these questions because they capture the organisation of feeding interactions that underpin energy flow, community structure, and ecosystem functioning (Cohen et al. 1990, Martinez and Dunne 1998, Dunne et al. 2002, Simmons et al. 2021). Documenting trophic interactions is therefore a fundamental component of community ecology and provides a powerful abstraction for mathematical and statistical models used to understand biodiversity and anticipate the consequences of environmental change (Windsor et al. 2023, Galiana and Araújo 2026).

13 Despite their widespread use, there is growing recognition that different food web repre-
14 sentations encode different assumptions about ecological processes and therefore support
15 different forms of inference. These challenges have been recognised since early discussions
16 of sampling effort, aggregation, and scale in food webs (Cohen et al. 1990, Martinez 1991,
17 Polis 1991, Blüthgen 2010, Dormann 2023), and remain central to debates about how
18 networks should be interpreted and applied (Blüthgen and Staab 2024). A key challenge
19 is therefore not simply how to construct food webs, but understanding what different
20 representations actually represent, which ecological processes they capture, and which
21 questions they can legitimately answer. A key challenge is therefore not simply how to
22 construct food webs, but understanding what different representations actually represent,
23 which ecological processes they capture, and which questions they can legitimately answer.
24 In this perspective, we provide an overview of alternative food web representations, fo-
25 cusing on how each embeds assumptions about the processes that determine interactions
26 (Section 3), the levels of organisation at which those processes operate (*i.e.*, biological,
27 ecological, spatial, and temporal scales), and the approaches used to construct the resulting
28 networks (Section 4).

29 Throughout this review we focus explicitly on trophic ecological networks (food webs).
30 Food webs provide a well-developed theoretical foundation for distinguishing interactions
31 that are evolutionarily feasible from those that are realised through ecological processes
32 such as spatiotemporal overlap, abundance, and consumer diet choice. Although ecological
33 communities are also structured by mutualistic, competitive, parasitic, and facilitative
34 interactions, we do not attempt to develop a general framework for all ecological interaction
35 networks. Extending this hierarchy beyond trophic interactions would require additional
36 assumptions about how non-trophic interactions become feasible and realised, and remains
37 an important direction for future synthesis.

38 Network construction reflects both the data used and the ecological theories governing
39 species interactions. Although the challenges associated with collecting and observing
40 interaction data have been extensively discussed (Pimm 1984, Polis 1991, Martinez and
41 Dunne 1998, Pringle and Hutchinson 2020, Brimacombe et al. 2023, 2024, Moulatlet et al.
42 2024, Blüthgen and Staab 2024, Saberski et al. 2024), there remains no clear synthesis
43 linking food web representations to the assumptions and scale-dependent processes that
44 underpin their construction. Clarifying these assumptions is essential for improving
45 ecological inference and for predicting how multiple stressors affect biodiverse communities

46 and their functioning.

47 Recent work by Gauzens et al. (2025) synthesised trophic network representations
48 according to node resolution and link type, highlighting the advantages, limitations, and
49 methodological considerations associated with different network types. Here we address a
50 complementary, but distinct, question: **what ecological processes give rise to those**
51 **different representations?**

52 We frame food web construction as a hierarchical transition from interaction feasibility
53 to interaction realisation. Rather than classifying networks according to their structural
54 characteristics, we focus on the assumptions embedded in their construction (about
55 ecological entities, feeding interactions, and the processes that constrain or realise them)
56 and show how these assumptions determine the ecological scale, ecological meaning, and
57 inferential scope of a network.

58 This perspective provides a process-based framework for linking food web representation to
59 ecological inference. In the following sections we examine how nodes, edges, and interaction
60 processes define alternative food web representations, how different construction approaches
61 encode different ecological assumptions, and why matching network representation to
62 the ecological question is essential for interpreting food webs in biodiversity science and
63 environmental change research.

64 **2 Setting the Scene: The Not So Basics of Nodes and** 65 **Edges**

66 Ecological networks represent an ‘object’ from which inferences can be made and thus
67 serve multiple purposes. While many aspects of community structure can be analysed
68 without networks (*e.g.*, trait distributions or abundance patterns) networks provide a
69 formal framework for capturing the organisation of interactions among species. The study
70 of network structure and topology has a long history in ecology, rooted in early theory on
71 energy flow (Lindeman 1942, Odum 1968), and later extended to questions of robustness,
72 stability, and complexity (*e.g.*, (May 1972, Pimm 1984, Dunne et al. 2002, Brose et
73 al. 2006, Montoya et al. 2006)). More recent work has built on this foundation to link
74 network structure to ecosystem functioning, persistence, and dynamical behaviour (*e.g.*,
75 (Schneider et al. 2016, Pilosof et al. 2017, Danet et al. 2024)). Networks are therefore
76 commonly treated as response variables in tests of ecological theory and statistical models

of the generative processes that give rise to observed structure and are widely used to compare communities across environmental gradients or through time (*e.g.*, Hao et al. 2025, Pecuchet et al. 2020). They also provide a platform for evaluating downstream responses to perturbations, including secondary extinctions and robustness to species loss (*e.g.*, (Dunne et al. 2002, Staniczenko et al. 2010, Keyes et al. 2021, 2024)), as well as inference about stability, ecosystem function, invasions, climate change, contaminants, and extinction cascades (*e.g.*, (Delmas et al. 2017, Curtsdotter et al. 2019, Terry et al. 2025)). Against this backdrop of multiple research agendas, the definition of ‘edges’ and ‘nodes’, and the levels of organisation at which they are defined, take many forms (Poisot et al. 2016b, Moulatlet et al. 2024), each of which encode a series of assumptions within a network. Here we introduce an alignment between these questions and baseline assumptions about nodes and edges.

2.1 How do we define a node?

While nodes are conventionally described as representing species, in practice they may correspond to a range of taxonomic and non-taxonomic units, including sub-species, genera, or families, as well as trophic species (*e.g.*, (Yodzis 1982, Williams and Martinez 2000)), feeding guilds (*e.g.*, (García-Callejas et al. 2023)), or life-stage-specific subsets of species (*e.g.*, (Clegg et al. 2018)). These choices reflect differences in the level, type, and consistency of resolution at which networks are constructed, rather than aggregation *per se*. Representing nodes at coarser or mixed resolutions can limit taxon-specific inference (*e.g.*, whether species *a* consumes species *b*), bias estimates of degree distributions (particularly generality and vulnerability) and complicate downstream analyses such as extinction or invasion dynamics, where species identity and the consequences of loss may be obscured (Beckerman et al. 2006, Clegg et al. 2018). At the same time, there are clear justifications for using aggregated representations when the distribution of interactions among functional or trophic units is more informative than species-level detail, for example when analysing extinction patterns across feeding guilds or other functional units (Dunhill et al. 2024). Issues of resolution, scale, and sampling have long been recognised as central to the construction and interpretation of food webs (*e.g.*, (Martinez and Dunne 1998, Dunne 2006)). Consequently, node definition is not merely a methodological choice but an assumption about the ecological entities represented in a network, with direct implications for the processes that can be represented and the questions that can be addressed (Gauzens et al. 2025).

2.2 What is captured by an edge?

Understanding how edge assumptions impact ecological process representation requires distinguishing between *potential* and *realised* links. Potential links reflect interaction feasibility, whereas realised links reflect ecological fluxes such as energy transfer. Links within food webs may therefore represent either potential interactions between species (Dunne 2006, Pringle 2020) or realised fluxes within a system, such as the transfer of energy or materials resulting from feeding interactions (Lindeman 1942, Proulx et al. 2005). Edges can thus correspond to different ecological ‘currencies’ (Gauzens et al. 2025). Moreover, links can be specified in multiple ways: they may be treated as present or absent (*i.e.*, binary), defined probabilistically (Poisot et al. 2016a, Banville et al. 2025), or represented by continuous functions that quantify interaction strength (Berlow et al. 2004). Link definition therefore depends on both the ecological currency and how interactions are represented. Because different edge definitions encode different ecological processes, they also support different forms of inference. For example, networks composed of feasible interactions can characterise potential trophic structure but are generally insufficient for inferring the magnitude or distribution of energy flow because their links are not environmentally or energetically constrained.

2.3 Network representations

These definitions of nodes and edges and associated assumptions give rise to two well established types of ecological networks: **metawebs**, which represent all *potential* interactions within a species pool (Dunne 2006), and **realised networks**, which represent the subset of interactions expressed within a particular community at a given place and time. Because of the assumptions made about nodes and edges in these networks, they differ in the scales they represent and the processes assumed to generate their structure.

A metaweb is fundamentally a list of *feasible* interactions between species. Metawebs identify evolutionarily and regionally plausible interactions, ecologically impossible (*i.e.*, forbidden) links (Jordano 2016a), and the potential diet breadth of species (Strydom et al. 2023). Feasibility is thus determined by trait complementarity, typically related to feeding, and can be further constrained by species co-occurrence, producing a transition from *global* to *regional* metawebs.

In contrast, realised networks are localised in space and time, with links shaped by co-occurrence, environmental conditions, and diet choice. Even when represented as binary

142 matrices, their links implicitly reflect interaction strength and energy transfer. Realised
143 networks are therefore not simply spatial or temporal subsets of metawebs; they emerge
144 from ecological processes that govern the occurrence and strength of interactions among
145 species. Consequently, a metaweb and realised network containing the same species may
146 differ substantially in structure because link presence is governed by different constraints.
147 Critically, links absent from a metaweb represent infeasible interactions, whereas links
148 absent from a realised network reflect context-dependent ecological constraints. At its most
149 general, a realised network must represent more than the potential for species interactions;
150 it must reflect the processes governing the distribution or strength of interactions within a
151 particular community.

152 **3 From Nodes and Edges to Process and Constraints**

153 In the previous section we discussed how the definition of nodes and edges, representing
154 different scales and processes, lead to the concept of a metaweb and a realised web. The
155 fundamental take-homes are that nodes vary in their resolution, edges vary in what kind
156 of process they represent and the intersection of these, defined by meta- vs. realised webs,
157 underpins distinct lines of inquiry and constraints on the type of inference we can make
158 with networks.

159 Following this, we develop a process-based framework describing the transition from
160 metawebs to realised food webs. Rather than treating network representations as alterna-
161 tive descriptions of the same community, we argue that they represent different stages in
162 the progression from interaction feasibility to interaction realisation. This transition is
163 governed by five principal trophic processes operating across evolutionary and ecological
164 scales: evolutionary compatibility, spatiotemporal overlap, abundance, consumer diet
165 choice, and non-trophic interaction modification Figure 1.

166 These processes are not intended as an exhaustive description of all determinants of
167 trophic interactions. Instead, they capture the principal assumptions embedded in the
168 food web representations discussed throughout this review and provide a framework for
169 understanding what different networks represent and what ecological inferences they
170 support.

171 [Figure 1 about here.]

3.1 Processes that determine the feasibility of an interaction

Evolutionary compatibility and co-occurrence are the two principal processes that define the feasibility of an interaction between two species. The scale of inference and set of processes embodied in these two constraints typically combine to define a ‘list’ of interactions that are viable/feasible and defined strictly as present/absent. Reflecting on the previous section, nodes are typically species and rules defining edges are defined by trait complementarity (phylogenetic) and/or co-occurrence. Here we provide more insight into each process.

Evolutionary compatibility

Interaction feasibility is first constrained by *functional compatibility* between consumers and resources. Traits associated with resource acquisition and defence determine whether a feeding interaction is physically and biologically possible, defining the set of feasible interactions within a metaweb. Evolutionary history and co-evolution shape these traits, making phylogenetic relationships and trait complementarity powerful predictors of interaction feasibility, rather than direct evidence that a particular interaction is realised.

This constraint is therefore rooted in the evolutionary history of consumers and resources (Rossberg et al. 2006, Gómez et al. 2010, Dalla Riva and Stouffer 2016, Segar et al. 2020), which manifests as trait complementarity between species (Benadi et al. 2022). In this body of theory, consumers possess the traits required to acquire and consume particular resources, while resources possess traits that permit or prevent consumption. Interactions that lack this compatibility are defined as *forbidden links* (Jordano 2016a); *i.e.*, feeding interactions that are physically or biologically impossible and therefore absent from the feasible interaction space represented by a metaweb.

Networks do not explicitly arise from models based on this constraint alone. Instead, evolutionary compatibility defines a set of feasible consumer-resource pairs, represented as binary (possible versus forbidden) or probabilistic interactions (Banville et al. 2025). For example, in the metaweb constructed by Strydom et al. (2022), probabilities quantify the confidence that a particular feeding interaction is possible between two species. A network constructed from evolutionary compatibility is therefore conceptually aligned with a *global metaweb*, representing the global feasibility of trophic links even among species that do not currently overlap in space or time (Figure 1).

Spatiotemporal overlap (encounter opportunity)

204 A feeding interaction can only be realised if consumers and resources overlap in space
 205 and time. We therefore distinguish *spatiotemporal overlap* (the ecological opportunity
 206 for species to encounter one another) from *observed co-occurrence*, which is a statistical
 207 pattern that may arise for many reasons unrelated to direct interactions (Blanchet et al.
 208 2020). Spatiotemporal overlap constrains which feasible interactions from a global metaweb
 209 can potentially occur within a particular region or community. It therefore provides the
 210 transition from global to regional metawebs without implying that all overlapping species
 211 *will* interact (Dansereau et al. 2024a). In the context of Figure 1 this would be the
 212 metawebs for regions one and two.

213 Spatiotemporal overlap is therefore a necessary, but not sufficient, condition for trophic
 214 interactions. A regional metaweb may be defined for a local community, but locality
 215 alone does not imply that interactions are realised; locality constrains the set of feasible
 216 interactions, whereas realised food webs represent interactions expressed under local
 217 ecological conditions.

218 We reinforce that a regional metaweb is a network whose edges represent *feasible* consumer-
 219 resource interactions under regional species overlap, rather than interactions that have
 220 been realised within a community. Although it is possible to visualise a network from
 221 the list of interactions generated by these constraints, it is important to be aware that
 222 the structure of this network is not constrained by any community context - just because
 223 species are able to interact does not mean that they will (Poisot et al. 2015, Caron et al.
 224 2024). This distinction is important because many empirical food webs are constructed
 225 from species occurrence or co-occurrence information. Confusing encounter opportunity
 226 with realised feeding interactions blurs the distinction between regional metawebs and
 227 realised food webs, and can lead to fundamentally different ecological interpretations of
 228 otherwise similar network structures.

229 **3.2 Processes that realise networks**

230 The previous processes define which trophic interactions are feasible. Here we focus on
 231 the ecological processes that determine which feasible interactions are realised within a
 232 community. Realised food webs emerge from the interaction between community context
 233 (spatiotemporal overlap, abundance, and non-trophic interactions) and individual consumer
 234 behaviour (diet choice), transforming feasible trophic interactions into realised patterns of
 235 energy transfer within ecological communities (Figure Figure 1).

236 These processes operate after evolutionary compatibility has constrained the feasible
237 interaction space. Rather than creating new trophic interactions, they filter, strengthen,
238 weaken, or prevent the realisation of feasible links under particular ecological conditions.

239 **Abundance**

240 Abundance is the simplest ecological process through which feasible trophic interactions
241 become realised. Under a null model of energy acquisition, consumers feed randomly among
242 feasible resources, consuming prey in proportion to their relative abundance (Stephens
243 and Krebs 1986). Abundance therefore provides a null expectation for the distribution
244 and strength of realised trophic links (Vázquez et al. 2009).

245 Importantly, abundance operates within the set of evolutionarily feasible interactions
246 defined by the metaweb. Species abundances influence encounter frequencies and the
247 probability that feasible feeding interactions are realised, but they do not determine
248 whether incompatible species can interact. Abundance data, linked to a regional metaweb,
249 therefore provides a foundation from which the distribution and strength of realised trophic
250 interactions can be inferred.

251 Abundance does not create trophic interactions independently of evolutionary compatibility.
252 Instead, among the set of feasible interactions, species abundances influence encounter
253 frequencies and the probability that feeding interactions are realised (Poisot et al. 2015,
254 Banville et al. 2025). Abundance therefore provides a null expectation for the distribution
255 and strength of realised trophic links, rather than determining whether incompatible
256 species can interact.

257 **Diet choice**

258 It is well established that consumers make more selective decisions than feeding on
259 resources purely in proportion to their abundance (Stephens and Krebs 1986). Consumer
260 diet choice is underpinned by energetic cost-benefit trade-offs centred on prey profitability
261 and constrained by traits associated with resource acquisition and consumption (Smith
262 et al. 2021, Wootton et al. 2023). Consequently, energetic constraints are invoked in a
263 variety of approaches for constructing realised food webs (e.g. Beckerman et al. 2006,
264 Pawar et al. 2012, Portalier et al. 2019, Cherif et al. 2024).

265 Unlike metaweb approaches, these models generate realised food webs as emergent outcomes
266 of consumer behaviour. We distinguish these approaches from models such as the Niche
267 Model (Williams and Martinez 2000), where diet choice is implicit within a probabilistic

network-generating function designed to reproduce expected network structure, rather than emerging from node-level ecological rules. We use *diet choice* as an umbrella term for processes derived from optimal foraging theory (Pyke 1984) and metabolic theory (Brown et al. 2004) that determine the energetic distribution and strength of realised trophic interactions.

Non-trophic interactions

We include non-trophic interactions not as trophic links in their own right, but as ecological processes that modify the realisation or strength of trophic interactions within realised food webs (Ings et al. 2009, Miele et al. 2019). They provide community context beyond evolutionary compatibility, spatiotemporal overlap, and abundance by altering encounter rates, feeding success, mortality, or access to resources (Golubski and Abrams 2011, Pilosof et al. 2017).

These effects may arise through direct interactions among consumers (e.g. predator interference or competition for space) or through indirect ecological interactions (e.g. refuge provisioning, habitat modification, or facilitation) that alter the probability or strength of trophic interactions (Ings et al. 2009, Staniczenko et al. 2010, Golubski and Abrams 2011, Kamaru et al. 2024).

Some interactions, such as pollination, illustrate why trophic and non-trophic processes are not always cleanly separable. Pollinators consume floral resources through trophic interactions while simultaneously influencing plant reproduction, recruitment, and population persistence through non-trophic processes (Holland et al. 2002, Bascompte and Jordano 2007, Sauve et al. 2016). We treat these systems as examples of how non-trophic processes modify trophic network realisation, rather than attempting to incorporate mutualistic networks into this framework (Kéfi et al. 2012, 2015, Miele et al. 2019, Buche et al. 2024).

4 Network construction

The above five processes are central to understanding the assumptions inherent in building different types of networks. Each of the processes, or combinations thereof, deliver a unique set of boundary conditions on what a network represents and can be used for. Here we build on the introduction of these five processes to further categorise the approaches to constructing networks ultimately moving to showcase how different construction approaches, encoding different assumptions about how interactions arise, can

299 support different ecological questions.

300 **4.1 Why construct food webs?**

301 Food webs are representations of biodiversity, but they are also models of ecological
302 systems. In an ideal world, every trophic interaction could be observed directly. In
303 practice, empirical interaction data are costly, incomplete, and often impossible to collect
304 across entire communities (Jordano 2016a, 2016b, Poisot et al. 2021). As a result, food
305 webs are constructed from a combination of empirical observations and ecological inference,
306 with different approaches embedding different assumptions about how interactions arise.

307 These assumptions matter because food webs support ecological inference beyond species
308 richness alone. Network topology underpins analyses of community structure, energy
309 flux, extinction cascades, ecosystem functioning, and dynamical stability (Danet et al.
310 2024). Consequently, interpreting downstream analyses requires understanding both the
311 evidence used to construct a food web and the ecological processes represented by its links
312 (Section 3).

313 Recent work has further highlighted that variation in food web topology can arise not
314 only from ecological differences among communities, but also from differences in sampling
315 strategy, taxonomic resolution, and network construction methodology (Brimacombe et al.
316 2023, 2024). Understanding how food webs are constructed is therefore a prerequisite for
317 understanding what ecological questions they can legitimately answer (Figure 2).

318 [Figure 2 about here.]

319 **4.2 Construction from empirical evidence**

320 Empirical food webs are constructed from observations of realised trophic interactions.
321 However, different empirical methods observe different components of feeding interactions
322 and therefore differ in their temporal resolution, taxonomic resolution, uncertainty, and
323 the ecological meaning of the resulting links.

324 Direct observations record feeding events as they occur in the field or laboratory and
325 provide the strongest evidence that an interaction has been realised under sampled
326 conditions (Jordano 2016a). These observations are, however, typically incomplete because
327 interactions are difficult to observe continuously across entire communities.

328 Gut and stomach content analyses reconstruct recently consumed prey items, while scat

analysis and DNA metabarcoding extend this approach by identifying prey DNA within faecal samples. These methods provide empirical evidence of realised feeding interactions but represent snapshots of recent diet and are subject to biases associated with digestion, detection, and sampling effort (Pompanon et al. 2012, Sousa et al. 2019).

Stable isotope analyses infer assimilated energy pathways and trophic position over longer temporal windows, providing indirect evidence for trophic interactions rather than direct observations of feeding events (Post 2002). Similarly, molecular approaches including environmental DNA and interaction metabarcoding can reveal components of trophic interactions that are otherwise difficult to observe, albeit with varying levels of certainty about the underlying interaction.

Collectively, these approaches construct realised food webs from empirical evidence, but each captures different aspects of trophic interactions. Consequently, empirical food webs should be interpreted in the context of the sampling method and evidence used to define their links.

4.3 Construction through ecological inference

When empirical interaction data are incomplete or unavailable, food webs can be reconstructed by inferring trophic interactions from ecological rules, species traits, phylogenetic relationships, or existing network structure. These approaches can be used to interpolate or gap-fill existing empirical datasets (e.g., Biton et al. 2024, Stock et al. 2017, Dallas et al. 2017, Poisot et al. 2023), reconstruct feasible interactions among species pairs, or generate expected network structure in the absence of species-specific interaction data. The resulting representations therefore range from predictions of individual links to complete metawebs or synthetic food webs, depending on the information and assumptions used.

A useful distinction among these approaches is whether inference is made from properties of the species involved, from the structure of an existing network, or from rules describing ecological processes. Here we first consider species-specific approaches, which use information about consumers and resources to infer interactions, before considering structure-based network generators, which generate network topology from assumptions about network structure rather than species-specific properties.

4.3.1 Species-specific inference

Trait- and species-based approaches infer trophic interactions from properties of consumers and resources, and therefore explicitly represent the evolutionary compatibility processes described in Section 3.1. These approaches assume that feeding interactions are governed by ecological rules, that these rules are reflected in species traits or phylogenetic relationships, and that trait compatibility can be used to infer feasible trophic interactions (Morales-Castilla et al. 2015, Dalla Riva and Stouffer 2016, Bramon Mora et al. 2018).

Species-specific approaches construct or reconstruct networks from expected feeding interactions between particular species pairs and are therefore closely associated with the construction of metawebs. These approaches generally rely on three assumptions: that feeding rules underpin interaction feasibility (Morales-Castilla et al. 2015); that some of these rules are conserved across species or phylogenetic lineages (Dalla Riva and Stouffer 2016, Bramon Mora et al. 2018); and that interactions can be inferred from properties of consumers and resources, including their functional traits.

Evolutionary compatibility and co-occurrence constraints have been critical to the construction of ‘first draft’ networks for communities for which we have no interaction data (Strydom et al. 2022). They are also central to interpolation in data poor regions and predicting interactions for ‘unobservable’ communities *e.g.*, prehistoric networks (Yeakel et al. 2014, Fricke et al. 2022, Dunhill et al. 2024) or future, novel community assemblages (Van der Putten et al. 2010). Furthermore, they have the capacity to evaluate a role of interactions among species relative to their distribution by accounting for the role of the environment and the role of species interactions (Albouy et al. 2014, Pollock et al. 2014, Gravel et al. 2019, Higinio et al. 2023).

Feeding rules can be specified in several ways. Expert-based approaches define feasible interactions using trait matching (Roopnarine 2017, Shaw et al. 2024), while mechanistic approaches often rely on trait-based relationships between consumers and resources (Rohr et al. 2010, Gravel et al. 2013). Statistical and machine-learning methods instead infer interaction probabilities from observed networks using ecological predictors such as traits and phylogeny (Pichler et al. 2020).

Rules are also defined by correlating real world interaction data with suitable ecological proxies for which data is more widely available (*e.g.*, traits) using some sort of binary classifier (see Pichler et al. (2020) for an overview). These include generalised linear models (*e.g.*, Caron et al. 2022), random forest (*e.g.*, Llewelyn et al. 2023), trait-based

391 k-NN (*e.g.*, Desjardins-Proulx et al. 2017), and Bayesian models (*e.g.*, Eklöf et al. 2013,
392 Cirtwill et al. 2019).

393 Finally, graph embedding uses the structural features of a known network to infer the
394 position of species in an unknown network through the decomposition of the interaction
395 onto the embedding space (see Strydom et al. (2023)). This decomposition relies on a
396 combination of ecological proxies (*e.g.*, phylogenetic relatedness Strydom et al. (2022)) in
397 conjunction with known interactions to infer the latent values of species, which can then
398 be mapped onto decomposition of a known network.

399 **4.3.2 Structure-based network generators**

400 Structure-based network generators construct food webs from probabilistic assumptions
401 about expected network structure rather than from properties of particular consumer-
402 resource pairs. They are commonly used as null expectations, comparative baselines, or
403 inputs to dynamical models. Because links are generated according to network-level rules,
404 rather than species-specific information, these approaches can generate complete synthetic
405 food webs even when little empirical information is available for the focal community.
406 However, the resulting networks represent expectations about food web structure rather
407 than predictions of which particular species interact.

408 The ability of these models to reproduce statistical properties of empirical food webs does
409 not necessarily imply that they recover the mechanisms generating particular interactions.
410 Their value therefore depends on whether the objective is to generate realistic network
411 structure as a null or comparative expectation, or to infer the trophic relationships of a
412 particular community.

413 Constructing feasible or realised networks can be framed as an ‘inductive reasoning’ process
414 where insight and generalisation arise from a set of observations and relationships around
415 feeding. Inductive reasoning as a foundation for network construction is implemented at
416 node and network levels.

417 These models generate ecologically realistic network structures using simple probabilistic
418 rules. They are commonly used as null expectations, for comparative analyses, and as
419 inputs to dynamical models. The determination of links between species is not directly
420 linked to properties of the nodes. This means these networks are usually not species specific.
421 Although they require little empirical information, they encode explicit assumptions about
422 expected network structure.

423 Stochastic network models use a probabilistic ruleset about diet choice and niche breadth
 424 to reflect fundamental ideas of foraging biology. These models that are based on the
 425 compartmentalisation and acquisition of energy for species at different trophic levels
 426 (Krause et al. 2003, Allesina and Pascual 2009) and that network structure can be
 427 determined by distributing interactions along single dimension (the ‘niche axis’; Allesina
 428 et al. (2008)). Typically, these models parametrise some aspect of the network structure
 429 (although see Allesina and Pascual (2009) for a parameter-free model). These models
 430 include the most used network generator, the Niche model (Williams and Martinez 2000),
 431 as well as the original Cascade model (Cohen et al. 1990) and the derived Nested hierarchy
 432 model (Cattin et al. 2004). Even though these networks are derived without any real-world
 433 data they are still able to recover the structure of empirical networks (Stouffer et al. 2005).
 434 These models often form the basis for dynamic models *e.g.*, the allometric trophic network
 435 (Brose et al. 2006, Schneider et al. 2016) and bioenergetic food web models (Delmas et al.
 436 2017).

437 **4.4 Construction from realised ecological processes**

438 These approaches predict realised food webs by modelling ecological processes that de-
 439 termine whether feasible trophic interactions are expressed within communities. Rather
 440 than inferring interaction feasibility, they operationalise the abundance, diet-choice, and
 441 non-trophic processes introduced in Section 3.2 to generate networks representing realised
 442 energy transfer. The resulting networks are widely used to study energy flux, extinction
 443 dynamics, and ecosystem functioning. They also provide the structural backbone for
 444 dynamical systems modelling to address questions about stability-structure-productivity-
 445 function relationships, secondary extinction dynamics, species invasion and climate change.
 446 There are two broad groups of models in this deductive category.

447 These models capture the *behaviour* of the nodes by explicitly considering the properties
 448 of the different species in the community. Which means that there is a degree of variance
 449 in which links are predicted between species unlike the more ‘static’ predictions made by
 450 inductive models. However, these networks are costly to construct in real world settings
 451 (requiring data about the entire community, as it is the behaviour of the system that
 452 determines the behaviour of the part) and lack the larger diet niche context afforded by
 453 metawebs.

4.4.1 Abundance-based models

Neutral models operationalise the abundance process by assuming that consumers interact randomly among feasible resources in proportion to prey abundance (Krishna et al. 2008, Canard et al. 2014). Here links are solely determined by the relative abundance of the different species in the community. Although unrealistic as a complete explanation, neutral models can be combined with inductive approaches to generate more localised predictions (Pomeranz et al. 2019).

4.4.2 Energetics- and behaviour-constrained models

A broader class of realised-network models predicts trophic interactions from energetic constraints on foraging behaviour, often using consumer-resource body-size relationships as proxies for prey profitability and feeding costs. Unlike trait-based feasibility models, these approaches determine which feasible interactions are realised by modelling energetic decision-making within communities.

There is a broader group of models that focus on determining interactions in terms of energetic constraints on diet breadth, often using the ratio of consumer-resource body size as a proxy for capturing the energetic constraints of feeding. Models such as those developed by Portalier et al. (2019) and Wootton et al. (2023) are similar to the mechanistic approaches discussed in Section 4.3, however instead of determining interactions based on mechanistic feasibility it is rather constrained by the energetic cost of predation. Note that although these models do not place any explicit constraints on the expected structure of the network, the links should still be considered as ‘realised’ owing to the energetic constraint placed on links. A different subset of diet models (*e.g.*, Beckerman et al. 2006, Petchey et al. 2008) use a diet choice approach, however like the stochastic network models they also embed assumptions on network structure. Thus, these models predict both interactions and network structure simultaneously, although they may benefit in being refined by more explicitly accounting for trait-based (*i.e.*, feasibility) parameterisation (Curtsdotter et al. 2019).

4.5 Construction choices shape ecological inference

The approaches above differ not only in the data they require, but also in the ecological assumptions encoded within their links. Consequently, alternative construction frameworks applied to the same species assemblage may generate food webs with different topology,

interaction strengths, or extinction dynamics. Network construction should therefore be viewed as an explicit modelling decision that shapes ecological inference, rather than a neutral preprocessing step. We return to these implications in Section 5.

5 Making Progress with Networks

The motivation to leverage network ecology in conservation ecology, environmental risk assessment and natural resource management stems from a shift away from species/population specific measures of the effects of stress and disturbance to community level metrics of these impacts. These metrics, such as resilience and more generally stability, ecosystem function and biodiversity, are natural properties of networks. This suggests that modern conservation, risk assessment and resource management would benefit from robust network tools to support decision making.

This is also true in the disciplines of ecology and environmental science and their focus on abundance, distribution, functions and services that biodiversity provides (Loreau and de Mazancourt 2013). Major questions remain unresolved, for example, about stability-diversity-productivity relationships, the drivers and consequences of extinctions and invasions, and the impacts of multiple stressors operating at multiple ecological scales (Simmons et al. 2021). A network approach to answering these types of questions specifically allows us to evaluate how environmental gradients and anthropogenic stress map through direct and indirect effects among species in a complex community and reveal fundamental patterns and understanding of processes in the natural world.

To effectively use networks to aid us in answering questions about conservation/risk assessment/management and core ecological theory, we need to be mindful that we are mapping the *correct* network representation to the question of interest (Gauzens et al. 2025). Notably, there are certain questions that cannot be answered using specific network representations as the scale of the question of interest is fundamentally misaligned with the process captured by a specific network representation (Section 3.1), the underlying data that is used to construct it (Section 4), or both.

Here we discuss and map the different network representations shown in Figure 1 to ‘appropriate’ research questions and agendas (see also Table 1). We also highlight some of the key methodological challenges that currently limit our conceptualisation of a ‘network’ and thus impact their effective practical application in real world settings.

Table 1: Showcasing some of the broader avenues of inquiry, specifically how they map to the different network representations. Additionally, we highlight some studies that address or present opening discussions around each research question. Superscripts at the research question indicate the strength of the current literature in addressing that research question: $\checkmark$ indicates area with strong foundational research, Δ partial/emerging areas of research, and $\times$ areas where research is weak/largely absent

Common			
Network Rep-	Inferential		
resentation	Pitfall	Appropriate Inference	Representative Studies
Global	Treating	How will novel	Gravel et al. (2016a); Dunne
Metaweb	feasible links	communities respond	et al. (2002)
	as evidence of	to <i>e.g.</i> , extinction,	
	realised	turnover, invasion and	
	feeding or	rewilding $\checkmark$	
	energy flow.		
		Diet-based	Rooney and McCann (2012);
		conservation focusing	Curtsdotter et al. (2011);
		not only on the target	McDonald-Madden et al.
		species but the species	(2016)
		it might depend on for	
		food resources Δ	
		Rewiring	Gilljam et al. (2015);
		capacity/potential of	Staniczenko et al. (2010); Su
		species by looking at	et al. (2024); Marjakangas et
		the <i>entire</i> diets of	al. (2025)
		species Δ	
		Eco-Evolutionary	Poisot et al. (2015);
		dynamics and how they	Baskerville et al. (2011)
		relate to the	
		conservation and	
		origination of feeding	
		strategies $\times$	

Common			
Network Rep- resentation	Inferential Pitfall	Appropriate Inference	Representative Studies
Regional Metawebs	Assuming co-occurrence implies trophic interaction	Applied use potential of questions highlighted for global metawebs at the management scale <i>e.g.</i> , a protected area Δ Refinement/extension of species distribution models by incorporating co-occurrence and species associations <i>e.g.</i> , predator and prey $\checkmark$	Albouy et al. (2014); Pellissier et al. (2018); Estrada and Bodin (2008) Araújo and Luoto (2007); Kissling et al. (2012)
Realised Webs	Assuming observed binary links quantify interaction strength or complete diets.	The allocation of multiple stressors across networks Δ Temperature threshold to community collapse Δ Extinction and persistence after harvesting/invasion/ex- tinction $\checkmark$ Stability-diversity- productivity-function $\checkmark$	Crain et al. (2008); Beauchesne et al. (2021) O’Gorman et al. (2019); Petchey et al. (2010) Allesina and Tang (2012); Yodzis (2001) Thébault and Fontaine (2010); Rooney and McCann (2012)

Network Representation	Common		
	Inferential	Appropriate Inference	Representative Studies
	Pitfall	Explicitly tying ecosystem level processes and nutrient flows to networks [×]	Moore et al. (2004)
		Meta communities and the idea of meta-network-communities ^Δ	Gravel et al. (2016b); Gilarranz et al. (2017)

5.1 Key Eco-Evo-Conservation Questions

5.1.1 Global Metawebs

Global metawebs are most appropriate for questions centred on interaction feasibility and potential diet breadth. They provide a platform for exploring hypothetical or novel communities under climate change (Hui and Richardson 2019), species invasions, reintroductions, and rewilding, and the potential rewiring capacity (Marjakangas et al. 2025). Because they focus on feasible rather than realised interactions, global metawebs are also well suited to studying eco-evolutionary dynamics and how evolutionary history, natural selection, and phenotypic plasticity shape interaction niches.

5.1.2 Regional Metawebs

Regional metawebs extend these questions by explicitly incorporating species co-occurrence and therefore provide a more management-relevant perspective. They can be used to refine species distribution models and projections of future community composition (García-Callejas et al. 2023, Hao et al. 2025). However, caution is required when comparing regional metaweb structure across space or environmental gradients, as observed differences may reflect species turnover (*e.g.*, β -diversity) rather than changes in interaction processes.

5.1.3 Realised networks

Realised networks are best suited to questions concerning how community and environmental context shape interaction structure and ecosystem functioning (Thuiller et al. 2024). They provide the appropriate framework for studying stability, resilience, biodiversity

dynamics, ecosystem functioning, extinction cascades, invasions, climate change impacts, and network rewiring through time. By capturing interactions that are actually occurring, realised networks allow investigation of how perturbations propagate through communities and influence persistence.

The increasing availability of long-term interaction datasets is expanding opportunities to address these questions (Danet et al. 2024, Wooster et al. 2024). However, empirical datasets often accumulate interactions across extended periods, potentially obscuring temporal variation in realised interactions (Polis 1991). Developing approaches that better reconcile empirical networks with realised community dynamics therefore remains an important challenge.

5.2 Key methodological challenges

As noted above, the three network types highlight longstanding methodological challenges that limit both the precision and accuracy of ecological inference. Here we briefly review these challenges and emerging opportunities to address them.

Understanding what empirical data represents: Robust inference requires understanding what constitutes an observed interaction, whether recorded directly (predation events) or indirectly (*e.g.*, gut contents or stable isotopes). Because empirical networks often accumulate observations across space and time, they may be conceptually closer to metawebs than realised networks.

The validation of network structure: Considerable progress has been made in assessing how well we are able to recover pairwise interactions (Strydom et al. 2021, Poisot 2023, Abramov et al. 2026), yet there remains no clear framework for evaluating their ability to recover network structure (Allesina et al. 2008, Tylianakis et al. 2010). This raises two related questions: what constitutes an appropriate benchmark, and which aspects of network recovery matter most? For metawebs, accurately identifying both present and forbidden links may be essential, whereas for realised webs it remains unclear whether recovering pairwise interactions, aggregate properties (*e.g.*, connectance), or both should be the primary objective.

Transitioning between metawebs and realised webs: Most approaches for modelling realised networks do not explicitly incorporate evolutionary constraints (although see Van De Walle et al. (2023); Wootton et al. (2023)). Progress will likely require either ensemble approaches (Terry and Lewis 2020, Becker et al. 2022) or methods for downscaling

metawebs into realised networks (e.g., Roopnarine 2006). However, any such transition must retain clarity about the meaning of links - structurally realistic networks do not necessarily represent realised prey choice or energy flow.

Developing frameworks that allow transitioning between metawebs and realised representations will also facilitate integration with metacommunity, metaecosystem, and ecosystem-level theory (Massol et al. 2011, Liu et al. 2025). Doing so requires that we develop an understanding of what the appropriate spatial and temporal scales of networks are (Fortin et al. 2021), and specifically how these relate to network type and the question of interest.

Making networks more tractable in applied spaces: The application of networks to conservation and management remains limited by difficulties in defining network boundaries, aligning them with management units, and interpreting network metrics in policy-relevant ways (Dansereau et al. 2024b). Addressing these challenges will require stronger links between network structure and ecosystem function, alongside careful matching of analytical tools and network representations to management objectives (Pellissier et al. 2018, O'Connor et al. 2025).

Taken together, these challenges highlight three overarching messages. (i) Network representations are inseparable from the data and assumptions used to construct them; (ii) validation and benchmarking must be aligned with the intended network type and research question; and (iii) greater conceptual clarity is needed when transitioning between metawebs and realised networks, particularly in applied contexts. Explicitly articulating these distinctions is essential if networks are to be used rigorously and transparently across scales.

6 Concluding Remarks

Ecological network representations provide alternative abstractions of biodiversity, each defined by specific assumptions regarding ecological entities, interactions, and processes. Consequently, the suitability of a given representation depends on the ecological questions being addressed and the processes that the available data can meaningfully capture. Recognising these differences is essential for interpreting network structure, evaluating the scope of inference, and avoiding implicit assumptions associated with conventional approaches. By linking network representations to their underlying ecological processes and data

requirements, the framework developed here provides a basis for systematically comparing alternative network constructions and identifying the contexts in which they are most informative. Establishing this standard is essential for preventing the misinterpretation of network data and for addressing foundational questions in biodiversity science.

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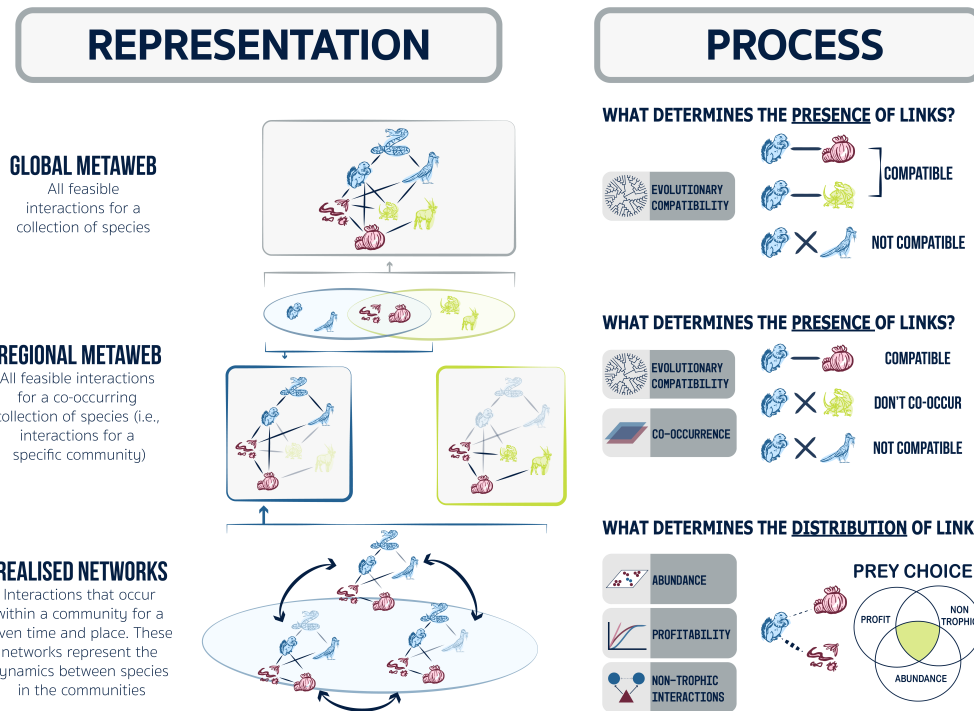


Figure 1: Aligning the processes that determine interactions (right) with different network representations (left). A **global metaweb** captures all feasible interactions among a species pool. Restricting this network by species co-occurrence yields **regional metawebs**, which differ in species composition. Species occurring in both regions are shown in red, whereas region-specific species are shown in blue and yellow. **Realised networks** represent subsets of regional metawebs expressed in specific spatial and temporal contexts. Their structure is shaped not only by co-occurrence, but also by community-level processes such as abundance, diet choice, and non-trophic interactions.

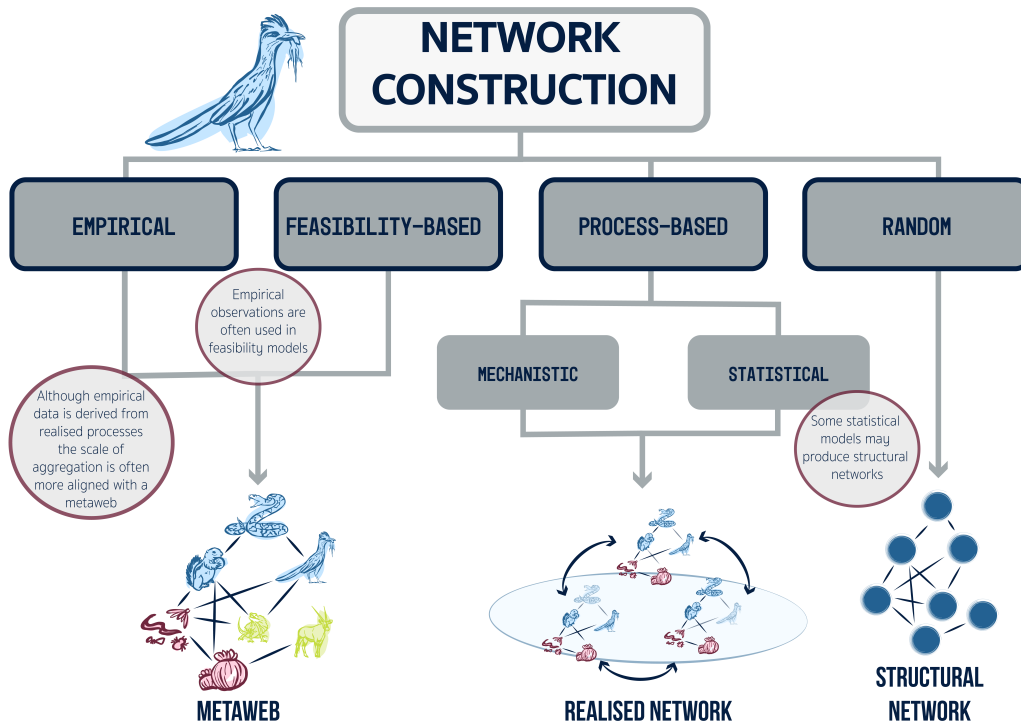


Figure 2: Flow chart showing the three core different approaches to building networks. Highlighting what the resulting network representation is.